BUSINESS REQUIREMENTS DOCUMENT

Mini ERP Project

Feature Summaries
Prepared by Kolosal AI

Version 1.0 · October 27, 2026 · Draft · Internal

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Product Detail Header

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Product Detail Header is the primary information hub located at the top of every product page in the Mini ERP system. It provides an immediate snapshot of a product's identity, current availability across different locations, and quick access to essential management tools. This feature ensures that staff can quickly identify a product and understand its stock status without needing to navigate away from the main view.

1.2 Step-by-Step Flow

1. Store Inventory: Displays the current stock levels available at retail store locations.
2. Warehouse Inventory: Displays the current stock levels held in warehouse facilities.
3. Stock History: Provides a direct link to view the historical logs of changes to store stock levels.
4. Product Identity: Displays the core product details, including the name, unique SKU, and current price.
5. Purchase History: Provides a link to view the full record of past customer purchases for the item.
6. Reservations: Shows the quantity of stock currently set aside for pending orders.
7. Warehouse History: Provides a direct link to view the historical logs of changes to warehouse stock levels.
8. In-Transit Status: Displays the quantity of stock currently being moved between locations.
9. Substitutes: Shows the number of alternative products available if the primary item is out of stock.
10. Product History: Provides a link to the audit log, which tracks historical changes to the product's details or pricing.

1.3 Key Concepts

- Inventory Visibility: Real-time tracking of stock across stores, warehouses, and transit, including items reserved for orders.
- Auditability: Built-in access to historical logs for stock movements and product attribute changes, ensuring transparency and accountability.
- Fallback Logic: Intelligent display settings, such as showing "N/A" for missing stock data or "Contact for pricing" when a price is not defined, to ensure the interface remains clean and informative.
- Operational Efficiency: Quick-access links to purchase and product histories, allowing users to perform administrative tasks directly from the product header.

1.4 Business Value

This feature centralizes critical product information, reducing the time staff spend searching for data across different screens. By providing immediate visibility into stock levels—including reservations and items in transit—the system helps prevent overselling and improves inventory accuracy. Furthermore, the inclusion of audit logs and

purchase history supports better decision-making for procurement and customer service, ultimately streamlining daily warehouse and retail operations.

BUSINESS REQUIREMENTS DOCUMENT

Product Search Bar Component

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Product Search Bar is a comprehensive filtering tool designed for the product page of our Mini ERP system. It allows users to quickly locate specific items by entering identifying details or selecting predefined categories. By providing a mix of text-based search fields and structured dropdown menus, the component ensures that users can perform both broad and highly specific product lookups with ease.

1.2 Step-by-Step Flow

1. Input Entry: As you type or select criteria in any of the fields (such as SKU, Name, Brand, or Tax rate), the system captures your choices in real-time.
2. Validation: If you enter information into a field that requires a specific format (like the SKU), the system will immediately alert you with a red highlight if the input is invalid.
3. Search Execution: Once you have entered your desired criteria, you click the "Apply Search" button.
4. Results: The system processes your selections and updates the product list to display only the items that match your criteria. If no filters are applied, the system defaults to showing all available products.

1.3 Key Concepts

- Filter Criteria: The collection of all search parameters (like Brand, Type, or SKU) that the system uses to narrow down the product list.
- Text Inputs: Flexible fields where you can type specific identifiers, such as a product name or code.
- Dropdown Selectors: Predefined lists that allow you to choose options like Brand or Tax rate, ensuring accuracy by preventing manual entry errors.
- Multi-select Tags: A compact feature that allows you to pick one or more descriptive labels to filter products by specific attributes.
- Visual Feedback: A safety feature that highlights fields in red if the information entered does not meet the required format, helping you correct errors before searching.

1.4 Business Value

This feature significantly improves operational efficiency by reducing the time staff spend manually scrolling through product lists. By offering a robust set of search filters, users can pinpoint exact inventory items in seconds. The intuitive design—combining free-text search for quick lookups with structured dropdowns for standardized data—minimizes user error and ensures that the search process is both fast and reliable.

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Products Sub Menu

Prepared by Kolosal AI

Version	Date	Status	Classification
1.0	October 27, 2026	Draft	Internal

1. Frontend

1.1 Overview

The Products Sub Menu is a navigation component designed to help users easily switch between different categories of product management within the ERP system. It functions as an expandable menu (accordion) that sits under the main Products section, ensuring that users can quickly access either the general Product List or the Service Products List with a single click.

1.2 Step-by-Step Flow

1. When a user accesses the Products area, the system automatically defaults to the "Products" view.
2. The user interacts with the sub-menu, which displays two primary options: "Products" and "Service Products."
3. Upon clicking "Products," the system updates the active selection and directs the user to the main Product List page.
4. Upon clicking "Service Products," the system updates the active selection and directs the user to the dedicated Service Products List page.
5. The menu visually highlights the currently selected category so the user always knows which section they are viewing.

1.3 Key Concepts

- Accordion Menu: A space-saving navigation style that allows users to expand or collapse sections to keep the interface clean.
- Active Tab: A visual indicator that shows the user which page or category is currently selected.
- Navigation Routing: The mechanism that ensures clicking a menu item instantly takes the user to the correct corresponding page in the ERP.

1.4 Business Value

This feature improves the user experience by providing a clear, organized way to navigate the product catalog. By grouping standard products and service products together in a dedicated sub-menu, we reduce the time users spend searching for specific management tools, leading to a more efficient workflow and a more intuitive interface for managing inventory and services.

BUSINESS REQUIREMENTS DOCUMENT

Products Menu

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Products Menu is a navigation component designed to help users easily browse and access product-related information within the ERP system. It functions as an interactive, accordion-style menu that allows users to expand or collapse sub-categories, ensuring the interface remains clean while providing quick access to specific product pages.

1.2 Step-by-Step Flow

1. Initial State: When the user first accesses the system, the "Products" menu is active by default, with its sub-menu already expanded to show available options.
2. Interaction: The user can click on any item in the navigation list.
3. Selection: Clicking an item updates the system to show the corresponding page for that selection and marks it as the "active" item.
4. Accordion Toggle: If the selected item has sub-categories, a chevron icon appears next to the text. Clicking the item toggles the sub-menu open or closed, allowing the user to reveal or hide nested links as needed.
5. Automatic Cleanup: If the user clicks anywhere outside of an open sub-menu, the system automatically closes it to keep the workspace organized.

1.3 Key Concepts

- Accordion Menu: A design pattern where clicking a header expands or collapses a list of items, helping to save screen space.
- Active State: A visual indicator that highlights which menu item or page the user is currently viewing.
- Chevron: A small arrow icon used to indicate that a menu item can be expanded to show more options.
- Outside-Click Closing: A user-friendly feature that automatically hides a menu when the user clicks away from it, preventing the screen from becoming cluttered.

1.4 Business Value

This feature improves the user experience by providing a structured and intuitive way to navigate complex product data. By using an accordion layout, we reduce visual noise, allowing users to focus on the specific product information they need without being overwhelmed by a long list of options. The consistent "active" highlighting ensures users always know their current location within the system, leading to faster task completion and reduced navigation errors.

BUSINESS REQUIREMENTS DOCUMENT

Main Menu

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Main Menu is the primary navigation header for the ERP system. It acts as the central hub for users to move between different areas of the software, manage their personal account settings, view system alerts, and quickly find specific information. The design ensures a clean interface by using dropdown menus for secondary actions, keeping the workspace uncluttered.

1.2 Step-by-Step Flow

1. Product Navigation: Clicking the "Products" item opens a sub-menu, allowing users to access specific product-related categories.
2. Profile Management: Clicking the user avatar toggles a menu for account settings and sign-out options. If other menus (like notifications) are open, they will automatically close to keep the interface tidy.
3. Alerts: Clicking the notification bell icon reveals a list of unread alerts. Similar to the profile menu, this will close other open dropdowns to prevent overlap.
4. Global Search: Users can type into the search bar to find information. To ensure smooth performance, the system waits for the user to finish typing (at least 3 characters) before displaying results.
5. General Navigation: Clicking any main navigation item updates the active tab to show the user where they are and redirects them to the corresponding page.
6. Sales Access: A dedicated button provides a direct shortcut to the Sales list page.

1.3 Key Concepts

- Dropdown Menus: Interactive containers that appear only when needed to keep the header clean.
- Active Tab State: A visual indicator that highlights the user's current location within the system.
- Debounced Search: A performance feature that waits for a brief pause in typing before searching, preventing unnecessary system load.
- Smart Toggling: A behavior where opening one menu automatically closes others, ensuring the user is never overwhelmed by multiple open windows.

1.4 Business Value

- Improved Efficiency: By providing a centralized navigation bar and quick-access shortcuts (like the Sales page), users can move through the ERP system with fewer clicks.
- Enhanced User Experience: The use of smart, self-closing menus and a responsive search bar creates a professional, intuitive environment that reduces user frustration.
- System Clarity: The "Active Tab" feature ensures users always know which section of the ERP they are currently working in, reducing navigation errors.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id/history

Prepared by Kolosal AI

Version	Date	Status	Classification
1.0	October 27, 2026	Draft	Internal

1. Data

1.1 Overview

The Product History feature allows users to view a detailed timeline of purchase prices for a specific product. By retrieving historical data, the system organizes past transactions by vendor, enabling you to monitor price fluctuations and vendor performance over time.

1.2 Step-by-Step Flow

1. Request Initiation: The system receives a request for a specific product's history, including pagination settings (which page to view and how many records to show).
2. Validation: The system verifies that the requested product exists in the database. If the product cannot be found, the request is rejected with a "Not Found" error.
3. Data Retrieval: If the product is valid, the system fetches the relevant purchase records from the database, sorted from the most recent to the oldest.
4. Data Formatting: The raw records are organized and grouped by vendor name, making it easy to see price trends per supplier.
5. Response: The system returns the structured, vendor-grouped price history to the user.

1.3 Key Concepts

- Pagination: A method of breaking large amounts of data into smaller, manageable chunks (pages) to ensure the system remains fast and responsive.
- Vendor Grouping: The process of organizing historical price data so that all transactions associated with a specific supplier are displayed together.
- Error Handling: Standardized responses that inform the user if something goes wrong, such as providing an invalid request (400), lacking permission (401), requesting a non-existent product (404), or encountering a technical server issue (500).

1.4 Business Value

This feature provides transparency into your procurement process. By tracking how purchase prices change over time and comparing them across different vendors, your team can make more informed purchasing decisions, identify cost-saving opportunities, and better manage vendor relationships.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id/warehouse-stocks

Prepared by Kolosal AI

Version	Date	Status	Classification
1.0	October 27, 2026	Draft	Internal

1. Logic

1.1 Overview

This feature allows you to view the current stock levels of a specific product across your various warehouses. By providing a product ID, you can retrieve a detailed breakdown of where that item is stored, including quantities that are reserved or in transit. The system also provides a summary of the total quantity available across all locations, with options to filter results or paginate through large lists of warehouse data.

1.2 Step-by-Step Flow

1. Product Verification: The system first checks if the requested product exists in the database. If the product cannot be found, it returns an error message.
2. Filtering and Search: If you provide specific criteria—such as a warehouse name, ID, or warehouse code—the system narrows down the search to only show relevant stock locations.
3. Data Retrieval: The system gathers the stock information for the product, combining it with details about the warehouses where the items are stored.
4. Calculation: The system calculates the total quantity of the product across the filtered warehouses.
5. Response: The system returns a summary of the total quantity along with a detailed list of the stock records. If any issues occur during this process (such as invalid requests or server errors), the system provides a standardized error message.

1.3 Key Concepts

- Product ID: A unique identifier used to pinpoint the specific item you want to check.
- Warehouse Filtering: The ability to search for stock by specific warehouse identifiers, names, or codes to quickly find the information you need.
- Pagination: A method of breaking down large lists of data into smaller, manageable pages (using "page" and "limit" settings) to improve performance and readability.
- Stock Breakdown: Detailed information regarding the status of items, including reserved stock, items currently in transit, and total available quantities.

1.4 Business Value

This feature provides real-time visibility into inventory distribution, which is essential for effective supply chain management. By allowing users to quickly identify which warehouses hold specific products and in what quantities, the business can:

* Improve Order Fulfillment: Quickly determine which warehouse is best positioned to ship an order.

* Optimize Inventory Levels: Easily identify stock imbalances across different locations.

* Enhance Decision Making: Use accurate, up-to-date stock data to inform purchasing and distribution strategies.

2. Data

2.1 Overview

This feature allows users to view the current inventory levels for a specific product across all available warehouses. By requesting information for a particular item, the system retrieves a clear breakdown showing exactly how many units are stored at each warehouse location.

2.2 Step-by-Step Flow

1. System Initialization: The system establishes a secure connection to the inventory database to begin processing the request.
2. Product Verification: The system checks the database to confirm that the requested product exists. If the product cannot be found, the process stops.
3. Stock Retrieval: Once the product is verified, the system searches for all inventory records associated with that specific item.
4. Warehouse Enrichment: The system matches the inventory records with warehouse details to identify the name of each location where the product is stored.
5. Data Formatting: The system organizes the collected information into a clean, easy-to-read list that pairs each warehouse name with the corresponding quantity of the product on hand.

2.3 Key Concepts

- Product: The specific item you are inquiring about, identified by its unique record in the system.
- Warehouse: A storage facility where inventory is kept; the system links stock levels to these specific locations.
- Warehouse Stock: The quantity of a specific product currently held at a particular warehouse.
- Enrichment: The process of combining raw stock numbers with descriptive warehouse names so the final report is meaningful to the user.

2.4 Business Value

This feature provides real-time visibility into inventory distribution. By knowing exactly where stock is located, your team can make informed decisions regarding order fulfillment, stock transfers between locations, and replenishment planning, ultimately reducing the risk of overselling or stockouts.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/store-stock-histories

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

1. Logic

1.1 Overview

This feature provides a way to view the history of stock changes within your stores. It allows users to retrieve a detailed log of inventory adjustments, such as restocks or sales, and filter these records to find specific information quickly. The system ensures that only authorized users can access this data and provides the results in organized, manageable pages.

1.2 Step-by-Step Flow

1. Authentication: The system verifies that the user is logged in. If the user is not authenticated, access is denied.
2. Filter Application: The system accepts specific search criteria from the user, such as the store location, the type of transaction, the date range, or the person who recorded the change.
3. Data Retrieval: The system searches the stock history records based on the provided filters.
4. Pagination: To keep the information easy to read, the system breaks the results into pages, providing the current page number, the number of items per page, and the total count of records found.
5. Response: The system delivers the requested list of stock history records along with the pagination details. If an unexpected error occurs during this process, the system notifies the user that a server error has taken place.

1.3 Key Concepts

- Stock History: A chronological record of all inventory movements or adjustments made within a store.
- Filtering: The ability to narrow down a large list of records by specific criteria (e.g., looking only at "restock" transactions for a specific store within a specific week).
- Pagination: A method of displaying a large dataset in smaller, bite-sized chunks to improve performance and readability.
- Authentication: A security check that ensures only verified users can view sensitive business data.

1.4 Business Value

This feature provides transparency and accountability for inventory management. By allowing managers to track exactly when, why, and by whom stock levels were changed, the business can better audit inventory discrepancies, identify trends in stock movement, and ensure that store operations are running accurately. The ability to filter and paginate this data ensures that managers can quickly find the specific information they need without being overwhelmed by large volumes of data.

2. Data

2.1 Overview

This feature provides a way to retrieve a detailed history of stock changes for a specific store. By using this tool, users can audit inventory adjustments, track when stock levels were modified, and identify who performed those changes. The system allows for flexible searching and filtering to help users quickly find the specific transaction records they need.

2.2 Step-by-Step Flow

1. Request Submission: The user sends a request to view stock history, optionally including filters such as the specific store, the type of transaction, a date range, or the name of the person who made the change.
2. Filtering: The system processes these criteria to narrow down the search results to only the relevant records.
3. Data Retrieval: The system queries the database to collect all matching stock history entries.
4. Pagination: To ensure the system remains fast and easy to navigate, the results are returned in organized pages rather than one massive list.
5. Response: The system returns the requested list of records. If an unexpected issue occurs during this process, the system provides an error message to notify the user.

2.3 Key Concepts

- Stock History: A chronological record of every time a store's inventory quantity was increased or decreased.
- Filtering: The ability to refine search results based on specific criteria, such as a date range or a specific staff member who recorded the change.
- Pagination: A method of breaking large amounts of data into smaller, manageable chunks (pages) to improve performance and readability.
- Transaction Type: A label used to categorize why a stock change occurred (e.g., a sale, a restock, or a return).

2.4 Business Value

This feature enhances inventory transparency and accountability within the Mini ERP system. By maintaining a clear audit trail of stock movements, store managers can:

* Improve Accuracy: Quickly investigate discrepancies between recorded stock and physical inventory.

* Enhance Security: Monitor staff activity to ensure all stock adjustments are authorized and legitimate.

* Support Decision Making: Analyze historical trends in stock changes to better understand store performance and replenishment needs.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id/prices/by-warehouse

Prepared by Kolosal AI

Version	Date	Status	Classification
1.0	October 27, 2026	Draft	Internal

1. Logic

1.1 Overview

This feature allows authorized users to retrieve a list of specific prices for a single product, broken down by warehouse. Because companies often set different prices for the same item depending on the location (e.g., due to regional taxes, shipping costs, or local market strategy), this tool provides visibility into those variations.

1.2 Step-by-Step Flow

1. Validation: The system checks that the provided product identifier is formatted correctly.
2. Security Check: The system verifies that the user is logged in and has the necessary permissions to view pricing information.
3. Product Verification: The system confirms that the requested product exists in the database.
4. Pricing Lookup: The system searches for all price records associated with that specific product across different warehouses.
5. Data Delivery: If successful, the system returns a list showing the warehouse location, the currency used, and the specific price for that location.
6. Error Handling: If any step fails—such as an invalid ID, lack of permission, or a system error—the user receives a clear message explaining why the request could not be completed.

1.3 Key Concepts

- Warehouse Price Override: A specific price assigned to a product that differs from the standard price, applied only when that product is sold from or stored in a particular warehouse.
- Product Identifier: A unique code used to ensure the system is looking up the correct item in the catalog.
- Authorization: The security process that ensures only users with the appropriate "view pricing" role can access sensitive financial data.

1.4 Business Value

This feature provides transparency into regional pricing strategies. By allowing staff to quickly pull a list of warehouse-specific prices, the business can:

- Improve Accuracy: Ensure that sales and inventory teams are using the correct pricing for specific regions.
- Support Decision Making: Help managers analyze how pricing varies across the supply chain to optimize profitability.
- Maintain Security: Protect sensitive pricing data by ensuring only authorized personnel can access these details.

2. Data

2.1 Overview

This feature allows the system to retrieve a list of specific prices for a product based on the warehouse where it is stored. In many business scenarios, the cost or sale price of an item may vary depending on its location; this tool provides a clear view of those "price overrides" so you can see exactly what a product costs at each individual warehouse.

2.2 Step-by-Step Flow

1. Request Initiation: The system receives a request to look up pricing for a specific product.
2. Validation: The system checks that the product identifier provided is valid. If the format is incorrect, the system returns a "Bad Request" error.
3. Authorization Check: The system verifies that the user has the necessary permissions to view sensitive pricing information. If not, it returns an "Unauthorized" or "Forbidden" error.
4. Data Retrieval: The system searches the database for all pricing records associated with that specific product.
5. Response Delivery:

* If records are found, the system returns a list showing the warehouse location, the currency, and the specific price for each entry.

* If no pricing records exist for that product, the system returns a "Not Found" error.

6. Error Handling: If an unexpected technical issue occurs during this process, the system returns a general "Internal Server Error" to ensure the system remains stable.

2.3 Key Concepts

- Price Override: A specific price assigned to a product that differs from the standard price, usually applied because of the product's location in a specific warehouse.
- Warehouse-Specific Pricing: The ability to maintain and view different price points for the same item depending on where it is stocked or sold.
- Validation: The process of ensuring that the information provided by the user is in the correct format before the system attempts to process it.
- Authorization: A security step that ensures only users with the correct access rights can view financial or pricing data.

2.4 Business Value

This feature provides transparency into your inventory costs and pricing strategy. By enabling the system to track and display price variations by warehouse, your team can:

* Improve Accuracy: Ensure that sales and procurement teams are looking at the correct pricing for specific regions or facilities.

* Support Dynamic Pricing: Easily manage and audit price differences caused by logistics, regional taxes, or local market conditions.

* Streamline Operations: Reduce manual errors by having a single, reliable source of truth for warehouse-specific pricing that is accessible via the system.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id/settings

Prepared by Kolosal AI

Version	Date	Status	Classification
1.0	October 27, 2026	Draft	Internal

1. Logic

1.1 Overview

This feature provides a dedicated way to manage specific "behavioral settings" for individual products within the Mini ERP system. Instead of bundling these settings with general product information, this feature uses a focused set of controls—such as serial number requirements, batch tracking, and consignment status—to dictate how the system handles a specific product during daily operations.

1.2 Step-by-Step Flow

1. Retrieving Settings: When you request the settings for a specific product, the system fetches the current status of nine key behavioral flags (e.g., whether the product requires serial numbers or batch tracking).
2. Updating Settings: When you submit changes to these settings, the system first verifies that your company has the necessary permissions to modify these features.
3. Validation: The system performs a series of safety checks to ensure the new settings are compatible with the product's current state. For example, it checks if a product already has existing transactions or stock before allowing you to toggle features like serial number tracking or consignment status.
4. Error Reporting: If any requested changes conflict with the product's history or current configuration, the system will reject the update and provide a list of specific reasons why the change is not allowed.
5. Saving and Syncing: If all validations pass, the system saves the new settings and automatically updates any related data links (such as consignor information) to ensure everything remains in sync.

1.3 Key Concepts

- Feature Flags: Specific "on/off" switches that determine how the system treats a product (e.g., "require_serial" forces the system to ask for a serial number during a sale).
- Validation Logic: A safety mechanism that prevents users from changing product settings if that change would break existing data or violate business rules (e.g., you cannot turn on batch tracking if the product doesn't have batch number fields configured).
- Consignment Settings: Specialized flags that manage how products held on behalf of others are tracked, ensuring that consignor details and stock levels are correctly accounted for.
- System Syncing: An automated background process that ensures that when a setting is changed, all related records in the system are updated to match the new configuration.

1.4 Business Value

This feature allows businesses to tailor the ERP system to the unique needs of different products without requiring custom development. By providing strict validation, it protects the integrity of your inventory data—preventing users

from accidentally enabling settings that would conflict with existing sales history or stock records. This ensures that your operational workflows remain consistent and accurate as your product catalog grows.

2. Data

2.1 Overview

This feature provides a way to retrieve the specific behavioral settings for any given product in the system. These settings act as "feature flags" that determine how the system handles a product—for example, whether it requires a serial number, if it is part of a consignment, or if it is subject to taxes. By accessing this information, the system can ensure that each product behaves according to its unique configuration.

2.2 Step-by-Step Flow

1. The system receives a request to view the settings for a specific product.
2. The system searches the database for a saved configuration associated with that product.
3. If a configuration is found, the system returns those specific settings.
4. If no configuration has been set for that product yet, the system automatically provides a default set of settings where all options are turned off (set to "false").

2.3 Key Concepts

- Feature Flags: On/off switches that control specific product behaviors, such as tracking batches or enabling multiple pricing tiers.
- Default State: A safety mechanism that ensures the system remains functional even if a product hasn't been explicitly configured yet.
- Product Settings: A collection of rules that define how the ERP system should process a specific item during daily operations.

2.4 Business Value

This feature allows for granular control over product management. By centralizing these settings, the business can easily toggle specific requirements (like tax applicability or serial number tracking) on a per-product basis. This flexibility ensures that the ERP system can adapt to different types of inventory—such as consignment goods versus standard retail items—without requiring custom code changes for every new product type.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products?utf8= &q[attribute_predicate]=value&...&page_rows=

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Logic

1.1 Overview

This feature provides a flexible search and filtering tool for the product catalog within the Mini ERP system. It allows users to retrieve a list of products by applying various criteria—such as name, SKU, brand, or product type—and control the number of results returned.

1.2 Step-by-Step Flow

1. Security Verification: The system first checks if the user is logged in and has the necessary permissions to view the product list. If not, access is denied.
2. Input Validation: The system reviews the search criteria provided by the user to ensure the data is formatted correctly and follows the expected rules.
3. Data Lookup: The system searches the product database using the provided filters (e.g., matching a specific brand or product code). It also verifies that any specific categories or tags requested actually exist in the system.
4. Result Delivery: If the search is successful, the system returns the matching list of products. If the search encounters issues—such as invalid data, missing resources, or excessive request volume—it provides an appropriate error message.

1.3 Key Concepts

- Filtering: The ability to narrow down a large list of products by specific attributes like SKU, name, brand, or tax category.
- Pagination: The ability to limit the number of results returned in a single search, which helps keep the system fast and easy to manage.
- Authentication & Authorization: Security checks that ensure only verified users with the correct access rights can view product data.
- Validation: A quality-control step that ensures the search request is logical and formatted correctly before the system attempts to process it.

1.4 Business Value

This feature empowers users to quickly locate specific items within a large inventory without manual searching. By providing granular filtering options, it improves operational efficiency for staff managing product data, ensures that users only access information they are permitted to see, and maintains system stability by handling high volumes of requests gracefully.

2. Data

2.1 Overview

The Product Search API allows users to retrieve a list of products from the system by applying various filters. It is designed to be flexible, enabling users to search by specific product details (like name or SKU), categorize results by brand or type, and control the output format through sorting and pagination. The system includes robust error handling to ensure that requests are secure, valid, and performant.

2.2 Step-by-Step Flow

1. Request Submission: The user sends a search request containing specific criteria (such as product name, brand, or tags) and formatting preferences (such as how many results to show).
2. Validation: The system checks the request for common issues, such as missing authentication, incorrect formatting, or insufficient user permissions.
3. Filter Verification: The system confirms that any specific IDs provided in the search (e.g., a specific Brand ID or Tax ID) actually exist in the database.
4. Data Retrieval: If the request is valid, the system queries the database using the provided filters, applies the requested sorting, and limits the number of results returned.
5. Response: The system returns the matching product data or provides a clear error message if the request could not be processed.

2.3 Key Concepts

- Search Predicates: Specific criteria used to narrow down results, such as matching a partial SKU, a product name, or filtering by specific categories like Brand or Product Type.
- Pagination: A feature that limits the number of products returned in a single request, which helps keep the system fast and responsive.
- Sorting: The ability to organize the search results in a specific order (e.g., by name or date).
- Error Handling: A set of rules that ensures the system responds appropriately to issues, such as unauthorized access, invalid search parameters, or requests that exceed system limits.
- Data Integrity: The system verifies that any referenced categories (like Tags or Brands) are valid before attempting to search for products associated with them.

2.4 Business Value

This feature provides a powerful and efficient way for users to find exactly the products they need within the ERP system. By supporting granular filtering and pagination, it reduces the time spent searching through large

inventories and ensures that the application remains performant even as the product catalog grows. Furthermore, the built-in security and validation checks protect the system from unauthorized access and malformed data, ensuring a reliable experience for all users.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

1. Logic

1.1 Overview

This feature allows the system to retrieve detailed information for a specific product using its unique identifier. When a request is made, the system verifies the user's identity and the validity of the request before returning the product's core details, such as its name, stock-keeping unit (SKU), price, and creation date.

1.2 Step-by-Step Flow

1. Validation: The system checks if the provided product ID is formatted correctly. If the ID is invalid, the request is rejected.
2. Authentication: The system verifies that the user is logged in and authorized to access the information. If credentials are missing or invalid, access is denied.
3. Data Retrieval: The system searches the database for a product that matches the provided ID.
4. Verification: If no product is found with that specific ID, the system informs the user that the requested resource does not exist.
5. Success: If the product is found, the system returns the full product details in a standardized format.
6. Error Handling: If an unexpected technical issue occurs during the process, the system provides a general error message to ensure the user is aware that something went wrong.

1.3 Key Concepts

- Unique Identifier: A specific, one-of-a-kind code assigned to every product in the system to ensure we retrieve the correct item.
- Authentication: The security process of confirming that the person requesting the data has the necessary permission to view it.
- Standardized Schema: A consistent structure for the information returned, ensuring that product details (like price and name) are always presented in the same predictable format.
- Error Handling: Built-in safeguards that provide clear feedback if a request fails, whether due to a bad ID, lack of permission, or a system-wide issue.

1.4 Business Value

This feature is essential for the day-to-day operations of the Mini ERP. It enables staff and integrated systems to quickly look up specific product information, ensuring that pricing, inventory tracking, and product management remain accurate and accessible. By providing a reliable way to fetch individual product data, the system supports better decision-making and smoother workflows across the organization.

2. Data

2.1 Overview

This feature provides a way for the system to retrieve the details of a specific product using its unique identifier. When a request is made, the system will either return the complete product information—including its name, stock-keeping unit (SKU), price, and creation date—or provide a clear explanation if the request cannot be completed due to an error.

2.2 Step-by-Step Flow

1. Authentication Check: The system first verifies that the user is authorized to access the information. If the user is not authorized, the system returns an "Unauthorized" error message.
2. System Health Check: The system ensures the internal database is functioning correctly. If a technical failure occurs during the process, the system returns an "Internal Server Error" message.
3. Data Retrieval: If the user is authorized and the system is healthy, the system searches for the product. If found, it returns the full product details, including the unique ID, name, SKU, price, and the date it was added to the system.
4. Missing Item Handling: If the system is functioning correctly but the specific product ID requested does not exist in the database, the system returns a "Not Found" error message.

2.3 Key Concepts

- Product Entity: The standard set of information stored for every item, consisting of a unique ID, name, SKU, price, and creation timestamp.
- Unauthorized Error: A response triggered when the system prevents access because the user does not have the necessary permissions.
- Not Found Error: A response triggered when the system is working properly, but the specific product requested does not exist in the inventory.
- Internal Server Error: A response triggered when an unexpected technical issue prevents the system from completing the request.

2.4 Business Value

This feature is essential for the day-to-day operations of the Mini ERP. It allows staff and integrated systems to quickly look up accurate product information, ensuring that pricing and inventory data are consistent across the business. By including standardized error messages, the system also helps users and developers quickly

identify and resolve issues, such as incorrect product IDs or access permission problems, reducing downtime and administrative frustration.

BUSINESS REQUIREMENTS DOCUMENT

PUT /api/products/:id

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Logic

1.1 Overview

This feature allows authorized users to perform a full update on an existing product within the Mini ERP system. When a user needs to change a product's details—such as its name, SKU, price, or stock quantity—this function ensures the request is secure, valid, and correctly recorded in the database.

1.2 Step-by-Step Flow

1. Security Check: The system first verifies that the user is logged in and has the necessary permissions to modify product information.
2. Data Validation: The system checks the incoming information to ensure all required fields (name, SKU, price, and stock quantity) are present and correctly formatted.
3. Product Lookup: The system searches the database for the specific product using its unique ID. If the product does not exist, the process stops and notifies the user.
4. Update Execution: Once the product is found and the data is verified, the system updates the product record with the new information and logs the current date as the "last updated" timestamp.
5. Confirmation: Upon a successful update, the system returns the updated product details to confirm the changes were saved.

1.3 Key Concepts

- Authentication: The process of confirming the user's identity to ensure they are a recognized member of the system.
- Authorization: The process of checking if a verified user has the specific "permission" required to perform an update.
- Validation: A quality-control step that ensures the data provided is complete and follows the expected format before it is saved.
- Full Update: A method where the system replaces the existing product details with the new set of information provided in the request.

1.4 Business Value

This feature provides a reliable and secure way to maintain accurate product records. By enforcing strict validation and permission checks, it prevents unauthorized changes and ensures that inventory data—such as pricing and stock levels—remains consistent and trustworthy for the rest of the ERP system.

2. Data

2.1 Overview

This feature enables the system to perform full updates on existing product records within the Mini ERP. When a user submits changes for a specific product, the system validates the request, ensures the user has the necessary permissions, and updates the product details (such as name, price, or stock levels) while automatically recording the time of the update.

2.2 Step-by-Step Flow

1. Search: The system looks for the specific product in the database using the provided ID.
2. Verification: If the product does not exist, the system logs an error and notifies the user that the item could not be found.
3. Update: If the product is found, the system applies the new information and updates the "last modified" timestamp.
4. Confirmation: Upon a successful update, the system logs the success for audit purposes.
5. Error Handling: If any issues occur during the process, the system logs specific details based on the type of failure:

* Database Errors: Logs technical failures if the system cannot save the changes.

* Access Violations: Logs a warning if a user tries to update a product they do not have permission to change.

* Validation Errors: Logs a warning if the information provided is incomplete or formatted incorrectly.

* Authentication Failures: Logs a warning if the user is not logged in or lacks the required credentials to perform updates.

2.3 Key Concepts

- Product Record: The digital entry containing details like the product name, unique identification code (SKU), price, and current stock quantity.
- Validation: A security and quality check that ensures the data being saved meets the required business rules before it is accepted.
- Audit Logging: The process of recording system events (like successes or errors) to help administrators track activity and troubleshoot issues.
- Unauthorized Access: A security measure that prevents users from modifying data they are not permitted to touch.

2.4 Business Value

This feature ensures that product information remains accurate and up-to-date, which is critical for inventory management and sales reporting. By including robust error logging and security checks, the system protects data integrity and provides clear feedback to administrators, allowing them to quickly identify and resolve issues related to unauthorized access or incorrect data entry.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

1. Logic

1.1 Overview

This feature provides a way to retrieve a list of all products stored in the Mini ERP system. To ensure the system remains fast and easy to navigate, the results are delivered in smaller, manageable chunks (pagination). This allows users to view specific segments of the product catalog without needing to load the entire database at once.

1.2 Step-by-Step Flow

1. Request Validation: The system checks the request to ensure the requested page number and the number of items per page are valid, positive numbers.
2. Security Check: The system verifies that the user is authorized to access the product data. If the user is not logged in or lacks proper credentials, the request is denied.
3. Data Retrieval: The system calculates which specific products to show based on the requested page and limit, then fetches that subset from the database. It also calculates the total number of products available in the system.
4. Response Delivery:

- If products are found, the system returns the requested list along with metadata (the current page, the limit used, and the total count of all products).
- If no products exist or an error occurs (such as invalid input or a server issue), the system provides a clear error message explaining why the request could not be completed.

1.3 Key Concepts

- Pagination: A method of breaking a large list of items into smaller, numbered pages to improve performance and readability.
- Query Parameters: Specific settings included in the request (like "page" or "limit") that allow the user to control exactly which portion of the data they want to see.
- Metadata: Additional information provided alongside the product list—such as the total number of items—that helps the user understand the scope of the data they are viewing.
- Authentication: A security gate that ensures only verified users can access sensitive business information.

1.4 Business Value

This feature enables efficient management of the product catalog. By using pagination, the application remains responsive even as the product database grows significantly over time. Providing the total count of products alongside the list allows for better user interface design, such as enabling "jump to page" functionality or displaying progress indicators, which improves the overall user experience for staff managing inventory.

2. Data

2.1 Overview

This feature provides a secure and efficient way to retrieve a list of all products stored in the Mini ERP system. It is designed to handle large amounts of data by allowing users to view products in smaller, manageable chunks (pagination). The system also includes robust error handling to ensure that users receive clear feedback if a request is invalid, unauthorized, or if a system error occurs.

2.2 Step-by-Step Flow

1. Validation Check: The system first checks if the request is formatted correctly. If the request parameters are malformed, it returns a "Bad Request" error.
2. Security Check: The system verifies the user's identity. If the user is not logged in or lacks the necessary permissions, it returns an "Unauthorized" error.
3. Data Retrieval: If the request is valid and authorized, the system fetches the requested page of products from the database, sorted by the date they were created.
4. Metadata Inclusion: Along with the product list, the system provides metadata, including the total number of products available, to help the user navigate the list.
5. Resource Verification: If the system cannot locate the requested data, it returns a "Not Found" error.
6. System Monitoring: If an unexpected technical issue occurs during the process, the system logs the error for the technical team and returns an "Internal Server Error" to the user.

2.3 Key Concepts

- Pagination: A method of breaking a large list of items into smaller pages (e.g., "page 1 of 10") to improve performance and readability.
- Metadata: Additional information provided with the product list, such as the total count of items, which helps the user understand the scope of the data they are viewing.
- Authentication: The security process of verifying that a user is who they claim to be before granting access to the product data.
- Error Handling: The system's ability to gracefully manage and report issues—such as bad requests or server problems—so the user knows exactly why a request might have failed.

2.4 Business Value

This feature serves as the foundation for inventory management within the Mini ERP. By providing a paginated, reliable, and secure way to view the product catalog, it enables staff to quickly browse and audit inventory

levels. The built-in error handling ensures that the system remains stable and provides helpful feedback, reducing frustration and improving the overall reliability of the ERP platform.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id/purchases

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

1. Logic

1.1 Overview

This feature provides a way to view the purchase history for a specific product directly on its detail page. By requesting this information, the system will retrieve a list of all recorded purchases associated with that product, including details like the date, quantity, and price. Users can also refine this list by filtering for specific suppliers, sorting the results, or limiting the number of records displayed.

1.2 Step-by-Step Flow

1. Security Check: The system verifies that the user is logged in and has the necessary permissions to view purchase records.
2. Input Validation: The system checks that the provided search criteria (such as supplier IDs or page settings) are formatted correctly.
3. Product Verification: The system confirms that the specific product requested actually exists in the database.
4. Data Retrieval: If all checks pass, the system gathers the relevant purchase records, applies any requested filters or sorting, and presents the data.
5. Error Handling: If any step fails—such as an invalid request, a missing product, or exceeding usage limits—the system provides a clear error message to explain the issue.

1.3 Key Concepts

- Authentication & Authorization: Security measures that ensure only verified users with the correct access rights can view sensitive purchase data.
- Filtering: The ability to narrow down the list of purchases by selecting a specific supplier.
- Sorting: The ability to organize the purchase list (e.g., by date or price) to make the data easier to read.
- Pagination: The ability to control how many purchase records appear at once, which helps keep the product detail page fast and easy to navigate.
- Validation: A safety check that ensures the information sent to the system is in the correct format before processing begins.

1.4 Business Value

This feature improves operational transparency by allowing staff to quickly audit the procurement history of any item. By providing on-demand access to purchase trends, supplier performance, and pricing history directly within the product view, teams can make more informed inventory and purchasing decisions without needing to run separate, complex reports.

2. Data

2.1 Overview

This feature provides a way to view the complete purchase history for a specific product directly within the Mini ERP system. By accessing this information on the product detail page, users can see exactly when items were purchased, in what quantities, at what price, and from which suppliers. The system includes built-in tools to filter these results by supplier and sort the list, making it easier to manage inventory and procurement data.

2.2 Step-by-Step Flow

1. Request Validation: When a user requests the purchase history for a product, the system first checks that the request is formatted correctly (e.g., ensuring the supplier ID and sorting preferences are valid).
2. Identity and Access Check: The system verifies that the user is authorized to view this data. If the user does not have permission or is not logged in, the request is blocked.
3. Product Verification: The system confirms that the requested product actually exists in the database. If the product ID is invalid, the system returns a "Not Found" error.
4. Data Retrieval: If all checks pass, the system searches the database for all purchase records associated with that specific product.
5. Applying Filters and Sorting: The system applies any requested filters (such as narrowing the list to a specific supplier) and organizes the data based on the user's sorting preferences.
6. Displaying Results: The system returns the final list of purchase records to the product detail page for the user to view.

2.3 Key Concepts

- Purchase History: A record of all procurement activities for a specific item, including dates, quantities, and costs.
- Filtering: The ability to narrow down a large list of purchases to show only those from a specific supplier.
- Sorting: The ability to arrange the purchase list in a specific order (e.g., by date or price) to help users find information faster.
- Validation: A security and quality check that ensures the system only processes requests that are correctly formatted and safe to execute.

2.4 Business Value

This feature improves operational transparency by giving procurement and inventory managers immediate access to historical pricing and supplier performance data. By being able to quickly filter and sort purchase history, staff can

make more informed decisions about reordering products, negotiating with suppliers, and tracking cost fluctuations over time, ultimately leading to better inventory control and cost management.

BUSINESS REQUIREMENTS DOCUMENT

PUT /api/products/:id (d53965a7-0cca-4c5d-bafa-34d01bbf1a8e)

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Logic

1.1 Overview

This feature allows authorized users to update the details of an existing product within the Mini ERP system. By providing a specific product ID and the new information (such as name, SKU, or price), the system will modify the existing record and confirm the changes.

1.2 Step-by-Step Flow

1. Authentication: The system verifies that the user is logged in and has a valid session.
2. Permission Check: The system confirms the user has the necessary rights to modify product information.
3. Data Validation: The system checks that the information provided in the update request is formatted correctly.
4. Product Lookup: The system searches the database to ensure the product ID provided actually exists.
5. Conflict Check: The system ensures the requested changes do not conflict with the current state of the product data.
6. Update Execution: If all checks pass, the system saves the new information to the database and records the time of the update.
7. Confirmation: The system returns the updated product details to the user to confirm the changes were successful.

1.3 Key Concepts

- Authentication: The process of verifying the identity of the user to ensure they are who they claim to be.
- Authorization: The process of checking if a verified user has the specific permission required to perform an update.
- Validation: A quality control step to ensure the data being sent to the system is complete and in the correct format.
- Conflict Check: A safety mechanism that prevents data errors by ensuring that an update doesn't clash with existing information or business rules.

1.4 Business Value

This feature ensures that product information remains accurate and up-to-date as business needs change. By including strict security and validation checks, it prevents unauthorized users from altering data and protects the integrity of the product catalog, ensuring that all team members are working with reliable, current information.

2. Data

2.1 Overview

This feature allows you to update specific details of an existing product in the Mini ERP system. By providing a unique product ID and the information you wish to change (such as the name, price, or SKU), the system will locate the record, apply the updates, and confirm the changes by returning only the information that was modified.

2.2 Step-by-Step Flow

1. Locate and Update: The system searches for the product using the provided ID and applies the requested changes to the database.
2. Timestamping: The system automatically updates the "last modified" date for the product record.
3. Filter Response: Once the update is successful, the system prepares a confirmation message that includes only the specific fields that were changed.
4. Error Handling: The system performs checks throughout the process to ensure:

* The product exists (if not, it reports that the product was not found).

* The data provided is valid and does not conflict with existing records (e.g., ensuring a new SKU isn't already in use).

* The user has the necessary permissions to perform the update.

* The system remains stable even if a connection error occurs.

2.3 Key Concepts

- Product ID: A unique identifier used to ensure the system updates the correct item in the database.
- Partial Update: The ability to change only specific pieces of information (like price) without needing to resend the entire product record.
- Data Integrity: Safeguards that prevent invalid data or duplicate identifiers (like SKUs) from being saved, ensuring the database remains accurate.
- Response Filtering: A feature that keeps the system's confirmation message concise by only showing the fields that were actually updated.

2.4 Business Value

This feature streamlines inventory management by allowing users to quickly and accurately modify product information. By supporting partial updates, it reduces the risk of accidental data overwrites and minimizes the amount of data that needs to be sent to the server. Furthermore, the built-in validation and error handling ensure that the product catalog remains consistent and reliable for all users.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/products/:id/stock

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

1. Logic

1.1 Overview

This feature provides a way to check the current inventory levels for a specific product across different store locations. By requesting information for a unique product ID, the system retrieves a breakdown of how many units are available at each store where that product is stocked. This is essential for ensuring accurate inventory visibility before a customer completes a purchase or modifies their shopping cart.

1.2 Step-by-Step Flow

1. Validation: The system checks that the provided product ID is formatted correctly. If the format is incorrect, the request is rejected.
2. Authentication: The system verifies that the user has the necessary permissions to access inventory data. If the user is not authorized, the request is rejected.
3. Product Verification: The system searches the database to confirm that the requested product actually exists. If the product cannot be found, the system reports that it does not exist.
4. Inventory Retrieval: Once the product is confirmed, the system gathers all stock records associated with that product ID.
5. Data Delivery: The system returns a list showing the quantity of the product available at each specific store location.
6. Error Handling: If an unexpected technical issue occurs during this process, the system provides a general error notification to ensure the user is aware that the request could not be completed.

1.3 Key Concepts

- Product ID: A unique identifier assigned to each item in the catalog, used to ensure the system retrieves data for the correct product.
- Stock Level: The specific count of items available for sale at a particular store location.
- Authentication: A security check that ensures only verified users or systems can access sensitive inventory information.
- Inventory Availability: The real-time status of product quantities, which helps prevent overselling or errors during the checkout process.

1.4 Business Value

This feature acts as a critical bridge between the product catalog and the sales process. By providing accurate, store-specific inventory data, it enables the business to:

* Improve Customer Experience: Customers can see if an item is available at their preferred store before they attempt to buy it.

* Reduce Operational Errors: It prevents the system from processing orders for items that are out of stock.

* Support Multi-Store Operations: It allows the business to manage and display inventory across multiple physical or virtual locations from a single source of truth.

2. Data

2.1 Overview

This feature provides a way to check the current inventory levels for a specific product across all store locations. By requesting information for a particular product, the system will return a breakdown of how many units are available at each store where that product is stocked.

2.2 Step-by-Step Flow

1. Format Validation: The system first checks that the product identifier provided is in the correct format. If the format is incorrect, the request is rejected.
2. Authentication Check: The system verifies that the user has the necessary permissions to access inventory data. If the user is not authorized, the request is denied.
3. Product Verification: The system confirms that the requested product actually exists in the product catalog. If the product cannot be found, the system reports that it does not exist.
4. Inventory Retrieval: If the product is valid and exists, the system searches the inventory records to find all entries associated with that product.
5. Data Delivery: The system returns a list showing the quantity of the product available at each specific store location. If no stock records are found for a valid product, an empty list is returned.
6. Error Handling: If a technical issue occurs (such as a database connection failure), the system logs the error internally and provides a generic notification that the request could not be completed.

2.3 Key Concepts

- Product Identifier: A unique code used to distinguish one specific item from all others in the catalog.
- Inventory Record: A data entry that links a specific product to a store location and indicates the quantity currently on hand.
- Validation: The process of ensuring that the information provided by the user is formatted correctly and refers to a real item before attempting to retrieve data.
-

Unauthorized Access: A security measure that prevents users without proper credentials from viewing sensitive inventory information.

2.4 Business Value

This feature enables real-time visibility into stock distribution. By allowing staff or integrated systems to quickly query inventory levels across different locations, the business can improve order fulfillment accuracy, prevent overselling, and make more informed decisions regarding stock transfers between stores.

BUSINESS REQUIREMENTS DOCUMENT

GET /api/stores

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Logic

1.1 Overview

This feature provides a simple way for authorized users to retrieve a complete list of all stores currently registered in the system. By calling this service, the application can pull essential store details—such as the store's name, location, and creation date—to display them within the ERP interface.

1.2 Step-by-Step Flow

1. Authentication Check: The system verifies that the person or application requesting the data is logged in and has a valid session. If the request is not authorized, the process stops, and an error is returned.
2. Data Retrieval: Once authorized, the system searches the database to collect all available store records.
3. Success Response: The system compiles the list of stores, including their unique IDs, names, locations, and creation dates, and sends this information back to the user.
4. Error Handling: If an unexpected issue occurs during the database search, the system catches the error and provides a notification that an internal server problem has occurred.

1.3 Key Concepts

- Authentication: A security gate that ensures only verified users can access sensitive business data.
- Store Record: A collection of information representing a single physical or digital store, containing its identity, name, and geographic details.
- API Response: The structured information sent back by the system, which can either be the requested list of stores or an error message if something goes wrong.

1.4 Business Value

This feature acts as a foundational building block for the Mini ERP system. By providing a reliable way to fetch store information, it enables other parts of the software to accurately reference store locations and details. This ensures that inventory, sales, and management reports are always linked to the correct store, supporting better operational visibility and data accuracy across the organization.

2. Data

2.1 Overview

This feature provides a way for the system to retrieve a complete list of all stores currently registered in the Mini ERP. It acts as a directory service, allowing authorized users or applications to view store details, including their names and locations, in a single request.

2.2 Step-by-Step Flow

1. Authentication Check: The system first verifies that the person or application making the request has a valid session. If the request is missing a token or the token is invalid, the process stops, and an "Unauthorized" error is returned.
2. Data Retrieval: Once authenticated, the system queries the database to collect all available store records.
3. Response Delivery: If the database query is successful, the system returns the full list of stores. If no stores are found, it returns an empty list.
4. Error Handling: If an unexpected technical issue occurs during the database search (such as a connection failure), the system logs the error internally and provides a generic "Server Error" message to ensure the system remains secure and stable.

2.3 Key Concepts

- Authentication: A security gate that ensures only verified users can access the store directory.
- Store Entity: The core piece of information being retrieved, which includes the store's unique identifier, name, location, and the date it was created.
- Unauthorized Error: A standard response triggered when a user attempts to access the list without proper credentials.
- Internal Server Error: A fallback response used when the system encounters an unexpected technical problem, protecting the system from exposing sensitive internal details.

2.4 Business Value

This feature is essential for operational visibility within the Mini ERP. By providing a reliable way to list all stores, it enables other parts of the system—and the business team—to easily reference store locations and manage organizational data. It ensures that store information is centralized, accessible, and protected by standard security protocols.

BUSINESS REQUIREMENTS DOCUMENT

Customer

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

The Customer feature is the central hub of our Mini ERP system. It is designed to maintain a comprehensive profile for every customer, linking together their personal information, vehicle details, membership status, billing preferences, and financial history. By centralizing this data, the system ensures that staff have a 360-degree view of each customer, enabling more efficient service and accurate record-keeping.

1.2 Step-by-Step Flow

1. Claim Tracking: Records individual service or sales claims associated with a customer.
2. Membership Management: Defines available membership programs and links customers to them to track points and status.
3. Financial Deposits: Manages customer deposit balances and logs every transaction movement.
4. Vehicle Registry: Tracks detailed vehicle information, including make, model, and technical specifications, linked directly to the owner.
5. Pricing & Billing: Stores custom product pricing for specific customers and maintains dedicated billing profiles.
6. Customer Master Data: The core profile that aggregates all the above information, including contact details, tax information, and internal notes.
7. Customer Grouping: Organizes customers into categories to help with reporting and management.
8. Store Association: Ensures every customer record is correctly linked to a specific store location to maintain data integrity across the organization.

1.3 Key Concepts

- Customer Profile: The primary record containing contact details, tax IDs, and internal settings for each individual or business.
- Referential Integrity: A safety rule that ensures a customer cannot be linked to a non-existent store, vehicle, or membership, preventing data errors.
- Denormalized Balance: A system-calculated total of a customer's deposits that updates automatically whenever a new transaction is recorded.
- Customer Grouping: A classification tool used to organize customers for easier filtering and business analysis.
- Vehicle Registry: A specialized module that tracks technical vehicle data (like VIN or engine numbers) to support service and sales operations.

1.4 Business Value

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Improved Customer Service: Staff can instantly access a customer's full history, including their vehicles, billing preferences, and membership status, reducing the time spent searching for information.

- Data Accuracy: By enforcing strict links between records (such as ensuring a customer belongs to a valid store), the system prevents "orphaned" data and keeps reports reliable.
- Operational Efficiency: Automated features—such as calculating deposit balances and tracking membership points—reduce manual data entry and minimize the risk of human error.
- Scalability: The modular design allows the business to easily add new features (like new membership types or vehicle categories) without disrupting existing customer data.

BUSINESS REQUIREMENTS DOCUMENT

Product

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

The Product feature serves as the central hub for managing inventory and catalog information within the Mini ERP system. It is designed to track not only the core details of a product (such as its name, brand, and type) but also its movement, pricing history, and availability across different physical locations like stores and warehouses.

1.2 Step-by-Step Flow

1. Classification: Products are organized by specific types and brands to ensure consistent categorization.
2. Store and Warehouse Setup: The system defines the physical locations (Stores and Warehouses) where stock can be held.
3. Inventory Tracking: The system monitors stock levels at these locations, including reserved or consigned items, and maintains a detailed history of stock changes (increases or decreases).
4. Pricing Management: The system tracks various price points for products, allowing for different pricing strategies based on the region, store, or customer group.
5. Product Lifecycle: Every product is assigned a unique identifier (SKU) and can be tagged for easier searching. The system maintains a comprehensive history of changes to product details and purchase records to ensure full auditability.

1.3 Key Concepts

- Product (SKU): The primary record for an item, containing its core identity, pricing, and status (e.g., active, frozen, or stock-tracked).
- Stock Management: The ability to view inventory levels across different physical locations, distinguishing between available, reserved, and consigned stock.
- Pricing Tracking: A flexible mechanism to record how prices change over time and vary depending on the sales channel or location.
- Audit History: Dedicated logs that capture changes to product information and purchase history, providing a clear trail of who updated what and when.
- Categorization: The use of Brands, Types, and Tags to group products logically, making it easier to manage large catalogs.

1.4 Business Value

- Operational Efficiency: By centralizing product data and stock levels, staff can quickly identify where items are located and how much is available, reducing stockouts or overstocking.
- Financial Accuracy: Detailed purchase and price tracking ensures that the business can accurately calculate costs, margins, and tax obligations.

- Scalability: The structured approach to brands, types, and tags allows the business to grow its product catalog without losing control over organization.
- Accountability: With built-in history tracking for both product changes and stock movements, the business gains full visibility into inventory discrepancies and pricing adjustments.

BUSINESS REQUIREMENTS DOCUMENT

Stock History

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Stock History feature provides a detailed audit trail of all inventory changes for a specific product. It allows users to track how and when stock levels have fluctuated across different store locations, helping to maintain accurate inventory records and accountability.

1.2 Step-by-Step Flow

1. Accessing the View: Users navigate to the "Stock History" tab within the product details section.
2. Initial Data Load: Upon opening the page, the system automatically retrieves the full history of stock movements for that product and populates the list. It also loads a list of available stores to populate the filter dropdowns.
3. Filtering Data: Users can narrow down the history by selecting specific criteria, such as a date range, a particular store, the type of transaction (e.g., restock or sale), or the user who recorded the change.
4. Refining Results: By clicking the "Search" button or clicking on column headers, the system updates the table to show only the relevant records, sorted as requested.
5. Reviewing Details: The table displays a comprehensive list of changes, including the date, the store involved, the quantity added or removed, any associated notes, and the staff member responsible for the entry.

1.3 Key Concepts

- Stock Movement: Any event that changes the quantity of a product, such as a new shipment arriving or a customer purchase.
- Audit Trail: A chronological record that shows exactly who made a change, when it happened, and why (via notes).
- Filter Controls: Tools that allow users to search through large amounts of data to find specific transactions quickly.
- Quantity Change: A visual indicator showing whether stock increased (positive number) or decreased (negative number) for a specific event.

1.4 Business Value

This feature enhances operational transparency by providing a clear history of inventory activity. It helps managers identify discrepancies, verify that stock levels match physical counts, and hold staff accountable for inventory adjustments. By offering robust filtering and sorting, it significantly reduces the time required to investigate stock-related issues or perform routine inventory audits.

2. Logic

2.1 Overview

The Stock History feature provides a transparent view of how a product's inventory levels have changed over time. By tracking every movement—such as additions, removals, or adjustments—this feature allows you to see exactly when, where, and why stock levels were modified, helping you maintain accurate inventory records across your different store locations.

2.2 Step-by-Step Flow

1. Initial Load: When you open the product history page, the system automatically retrieves a list of all available stores to populate a selection menu.
2. Data Retrieval: The system simultaneously pulls the complete stock history for the specific product you are viewing.
3. Display: The history is presented in a clear table, showing the date of the change, the store involved, the type of movement, the quantity adjusted, the reason for the change, and who performed the action.
4. Filtering: You can narrow down the list using a variety of filters, including specific stores, movement types, date ranges, or the person who recorded the change.
5. Refinement: Clicking the "Search" button or sorting by table headers updates the view instantly, allowing you to quickly find specific transactions or audit trails.

2.3 Key Concepts

- Stock Movement: Any event that changes the quantity of a product, such as a sale, a restock, or a manual adjustment.
- Audit Trail: A chronological record that shows who made a change to the stock and why, ensuring account-ability.
- Filtering: The ability to isolate specific data points (like a date range or a specific store) to make large amounts of history easier to analyze.
- Store Mapping: The process of linking inventory data to specific physical or virtual locations, ensuring you know exactly where the stock change occurred.

2.4 Business Value

- Improved Accuracy: By maintaining a detailed log of all stock changes, you can easily identify and resolve inventory discrepancies.
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Enhanced Accountability: Knowing exactly who performed a stock adjustment and why provides better oversight of warehouse and store operations.

- Better Decision Making: With clear visibility into historical trends, you can make more informed decisions regarding restocking schedules and inventory management strategies.
- Time Savings: The built-in search and filter tools allow your team to quickly locate specific transaction details without manually sifting through paper records or spreadsheets.

3. Data

3.1 Overview

The Stock History feature provides a clear, chronological view of how your inventory levels have changed over time. By tracking every adjustment made to a product's stock, this feature allows you to see exactly when, where, and why inventory levels were updated, ensuring full transparency across your business operations.

3.2 Step-by-Step Flow

1. The system retrieves a complete list of all your configured store locations to ensure you can identify where stock movements are taking place.
2. The system looks up the specific history for the product you are viewing, gathering every recorded change associated with that item.
3. The system displays the collected data, showing the amount of stock added or removed, the reason for the change, and the date it occurred.
4. If no history exists for a product or no stores are configured, the system will simply show an empty list rather than an error.

3.3 Key Concepts

- Stock History: A detailed log that records every increase or decrease in inventory for a specific product.
- Change Amount: The specific quantity of items that were added to or removed from your inventory.
- Reason: A descriptive note explaining why a stock adjustment occurred (e.g., a new shipment, a customer sale, or a manual correction).
- Store: The specific physical or digital location where the inventory is held and managed.

3.4 Business Value

This feature is essential for maintaining accurate inventory records and operational accountability. By providing a clear audit trail, it helps you identify trends in stock consumption, investigate discrepancies, and make informed

purchasing decisions. Ultimately, it reduces the time spent manually tracking inventory changes and helps prevent stockouts or overstocking by providing visibility into your supply chain movements.

BUSINESS REQUIREMENTS DOCUMENT

Purchase History

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Purchase History feature provides a centralized view of all procurement activity for a specific product. It allows users to track when a product was purchased, from which supplier, in what quantities, and at what price. Additionally, the feature includes tools for inventory management, such as generating QR codes and barcodes for the product.

1.2 Step-by-Step Flow

1. Automatic Loading: When you open the page, the system automatically identifies the product and retrieves its full purchase history.
2. Data Display: The purchase records are presented in a table showing the supplier, date, quantity, and unit price for each transaction.
3. Document Access: Each purchase record includes a document number that acts as a link, allowing you to click through to view specific delivery details.
4. Filtering: You can use the supplier dropdown menu to narrow down the list to a specific vendor, then click the "Search" button to refresh the table with the filtered results.
5. Sorting: You can organize the list by clicking on the column headers (such as Date or Supplier) to sort the records as needed.
6. Inventory Tools: At any time, you can use the dedicated buttons to open a modal window to view or print the product’s QR code or barcode.

1.3 Key Concepts

- Purchase Record: A historical entry detailing a specific transaction, including the supplier, date, quantity, and cost.
- Document Number: A unique reference ID for a delivery that serves as a clickable link to more granular information.
- Supplier Filter: A tool that allows you to isolate purchase history based on the vendor who provided the product.
- Inventory Identifiers: Digital representations (QR codes and barcodes) used for scanning and tracking products within the warehouse.

1.4 Business Value

This feature improves inventory transparency and vendor management by providing an audit trail for every product. By consolidating purchase data, users can quickly analyze pricing trends, verify supplier performance, and easily access delivery documentation. The inclusion of integrated QR and barcode tools streamlines warehouse operations by ensuring that product identification is always accessible directly from the history page.

2. Logic

2.1 Overview

The Purchase History feature provides a clear, chronological view of every time a specific product has been purchased. When you view a product's details, the system automatically retrieves and displays a complete list of past procurement records, allowing you to track supplier information, quantities, and pricing over time without manual searching.

2.2 Step-by-Step Flow

1. Automatic Trigger: When you open the product details page, the system automatically identifies the product you are viewing.
2. Data Retrieval: The system searches the database for all purchase records linked specifically to that product.
3. Sorting: The records are organized automatically, showing the most recent purchases at the top of the list.
4. Display: The information—including the supplier, date, quantity, and price—is populated into a table for easy review. If no previous purchases exist, the system will simply show an empty list.

2.3 Key Concepts

- Purchase Record: A historical entry containing details about a specific transaction, such as who supplied the item, when it was bought, how much was ordered, and the cost per unit.
- Chronological Sorting: The practice of ordering data by date so that the most recent activity is always the most visible.
- Product Association: The link between a specific item in your inventory and the history of transactions related to that item.

2.4 Business Value

This feature improves inventory transparency and decision-making by providing instant access to historical procurement data. By seeing exactly when and at what price a product was last purchased, your team can better manage supplier relationships, identify price trends, and ensure more accurate budgeting and forecasting for future orders.

3. Data

3.1 Overview

The Purchase History feature provides a clear, chronological view of all past procurement activities for a specific product. When viewing a product in the system, users can instantly see a detailed list of every time that item was purchased, including who supplied it, when it was bought, the quantity ordered, and the price paid. This ensures that procurement teams have immediate access to historical data to inform future purchasing decisions.

3.2 Step-by-Step Flow

1. The system identifies the specific product currently being viewed by the user.
2. The system automatically requests all associated purchase records for that product.
3. The system retrieves the relevant data, ensuring that the most recent purchases appear at the top of the list.
4. The retrieved information is displayed in a table format for the user to review.
5. If no previous purchases exist for the product, the system will display an empty list rather than an error.
6. If the system encounters an issue (such as an invalid product ID or a temporary connection problem), it will provide an appropriate notification to the user.

3.3 Key Concepts

- Purchase History: A comprehensive log of all past transactions associated with a specific item, used for tracking supply trends.
- Chronological Sorting: The automatic organization of records by date, ensuring the newest transactions are always visible first.
- Data Integrity: The system’s ability to ensure that every purchase record is accurately linked to the correct product, preventing data mismatches.

3.4 Business Value

This feature eliminates the need for manual record-keeping or searching through external files to understand a product's procurement background. By centralizing this information, the business can:

- Quickly verify past pricing to negotiate better deals with suppliers.
- Monitor supply consistency by reviewing historical quantities and supplier names.
- Improve inventory planning by having easy access to the frequency and timing of past orders.

BUSINESS REQUIREMENTS DOCUMENT

Warehouse Stocks

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Warehouse Stocks feature provides a centralized view of inventory levels across different warehouse locations for a specific product. It allows users to monitor stock availability, track incoming and reserved items, and manage minimum and maximum stock thresholds. The interface is designed for easy navigation within the broader product management system, including quick access to printing tools like QR codes and barcodes.

1.2 Step-by-Step Flow

1. Page Initialization: Upon loading the page, the system automatically retrieves and displays the current stock data for the selected product across all warehouses.
2. Filtering: Users can narrow down the list of warehouses using the filter bar (by warehouse name or code). The table updates only after the user clicks the "Search" button.
3. Data Review: The table displays detailed metrics for each location, including total quantity, available stock, reserved units, items in transit, and cost analysis.
4. Editing: Users can toggle "Edit Mode" to update the minimum and maximum stock quantity thresholds directly within the table.
5. Navigation & Printing: Users can navigate to other product-related pages via the main menu or tabs, or click the dedicated buttons to generate and print QR codes or barcodes for the product.

1.3 Key Concepts

- Available Quantity: The amount of stock currently on hand that is free to be sold or used.
- Reserved Quantity: Stock that has been committed to existing orders and is no longer available for new sales.
- In Transit: Inventory that is currently being moved or shipped to the warehouse but has not yet arrived.
- Min/Max Quantity: Thresholds set by the business to trigger reordering (Min) or to define storage capacity limits (Max).
- Bulk Edit Mode: A feature that enables users to quickly adjust stock thresholds across multiple warehouses at once.
- Cost Analysis: A breakdown showing the average unit cost and the total value of the stock held at each location.

1.4 Business Value

This feature improves inventory accuracy and operational efficiency by providing real-time visibility into stock distribution. By allowing users to view both "available" and "reserved" quantities, the business can prevent

overselling and better manage replenishment cycles. Furthermore, the ability to set and edit minimum and maximum stock levels directly in the interface helps maintain optimal inventory levels, reducing the risk of both stockouts and overstocking.

2. Logic

2.1 Overview

The Warehouse Stocks feature provides a comprehensive view of inventory levels for a specific product across all company warehouses. When you view a product's details, the system automatically retrieves and displays a breakdown of stock status, costs, and replenishment thresholds, allowing you to monitor inventory health at a glance.

2.2 Step-by-Step Flow

1. Trigger: The system initiates the data retrieval process automatically when you open the product details page.
2. Data Collection: The system gathers all stock records associated with that specific product, including details about the warehouses where the items are stored.
3. Data Processing: The system organizes the raw data into a clear format, calculating the total inventory quantity across all locations.
4. Display: The processed information is presented in a table, showing specific metrics for each warehouse alongside a summary of the total stock available.

2.3 Key Concepts

- Available Quantity: The number of items currently on hand that are ready to be sold or used.
- Reserved Quantity: Items that are already allocated to specific orders but have not yet left the warehouse.
- In Transit: Stock that is currently being moved between locations or is on its way from a supplier.
- Min/Max Quantity: The established safety thresholds for inventory levels to help you decide when to reorder stock.
- Average/Total Cost: Financial metrics that provide insight into the value of the inventory held in each warehouse.

2.4 Business Value

This feature improves inventory visibility and operational efficiency by:

* Centralized Oversight: Eliminating the need to check multiple locations manually by providing a single, consolidated view of stock levels.

* Informed Decision Making: Helping managers identify which warehouses are overstocked or understocked based on the defined minimum and maximum thresholds.

* Financial Clarity: Providing immediate access to the cost value of inventory, which assists in better budget management and financial reporting.

3. Data

3.1 Overview

The Warehouse Stocks feature provides a clear, real-time view of inventory levels for a specific product across all your warehouse locations. It allows you to see exactly how much stock is on hand, how much is currently reserved or in transit, and the associated financial costs, helping you make informed decisions about stock replenishment and distribution.

3.2 Step-by-Step Flow

1. The system receives a request to view stock information for a specific product.
2. The system searches the database to match that product with all associated warehouse records.
3. For every warehouse where the product is stored, the system retrieves detailed inventory metrics, including current stock, reserved items, and items currently being moved (in transit).
4. The system calculates the total quantity of the product across all locations.
5. The final report is generated, displaying a breakdown by warehouse alongside the overall total quantity.

3.3 Key Concepts

- Inventory Quantity: The total amount of a product physically present in a specific warehouse.
- Available Quantity: The stock that is currently ready to be sold or used, excluding reserved items.
- Reserved Quantity: Stock that has been set aside for specific customer orders or internal commitments.
- In Transit Quantity: Stock that has been ordered or transferred but has not yet arrived at the warehouse.
- Min/Max Quantity: The defined thresholds for stock levels used to trigger reordering or prevent overstocking.
- Average/Total Cost: The financial valuation of the stock held in a specific warehouse, calculated based on unit costs.

3.4 Business Value

This feature eliminates guesswork regarding inventory availability. By providing a centralized view of stock levels across multiple locations, it enables your team to:

- Improve order fulfillment accuracy by knowing exactly what is available versus what is reserved.

- Optimize warehouse space and capital by monitoring minimum and maximum stock thresholds.
- Gain better financial visibility into inventory assets through automated cost calculations.
- Reduce the risk of stockouts or overstocking by tracking items currently in transit.

BUSINESS REQUIREMENTS DOCUMENT

Product List Page

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Product List Page is a central hub within the Mini ERP system designed to help users efficiently manage and view their inventory. It provides a structured table view of all products, allowing users to search, filter, and sort data. The page is integrated into the global navigation system, ensuring a consistent experience while providing quick access to product details and the ability to add new items to the catalog.

1.2 Step-by-Step Flow

1. Initialization: When the page loads, the system automatically retrieves the product list and displays it in a table format, showing key details like SKU, Name, Brand, Code, Tax, and Price.
2. Navigation: Users can use the global menu to move throughout the ERP or use the "Add Product" button to jump directly to the product creation screen.
3. Filtering: Users can narrow down the list by entering specific criteria (such as SKU, Name, or Category) or selecting options from dropdown menus (such as Brand, Tax, or Product Type).
4. Executing Search: Once filters are set, clicking the "Apply Search" button updates the table to show only the items that match the selected criteria.
5. Sorting: Users can click on column headers to sort the list. The first click sorts items in ascending order, and a second click reverses the order to descending.
6. Viewing Details: By clicking on a product name or code within the table, the user is redirected to a dedicated page containing the full details for that specific item.

1.3 Key Concepts

- Filter Bar: A set of input fields and dropdowns that allow users to refine the product list based on specific attributes like brand, tax, or product type.
- Sortable Columns: Interactive table headers that allow users to organize data alphabetically or numerically by clicking the sort icon.
- Row Selection: The ability to click on a specific product row to navigate to its detailed information page.
- Global Navigation: The main menu integration that ensures the user can always access other areas of the ERP system.
- Sub-Menu: A secondary navigation area that highlights the current section (Products) and helps users orient themselves within the module.

1.4 Business Value

This feature streamlines inventory management by reducing the time spent searching for specific items. By providing robust filtering and sorting capabilities, users can quickly locate products, verify pricing and tax information, and manage their catalog with minimal effort. The intuitive navigation ensures that moving between the product list and individual product details is seamless, supporting a more productive and error-free workflow for the business.

2. Logic

2.1 Overview

The Product List Page feature provides a centralized interface for users to view, search, and manage the company's product catalog. It allows users to quickly locate specific items using search criteria and browse through the inventory in an organized, paginated format. This ensures that even with a large number of products, the system remains fast and easy to navigate.

2.2 Step-by-Step Flow

1. Initial Load: When you open the Product List Page, the system automatically retrieves the first set of products from the database.
2. Search and Filter: You can enter specific keywords (such as a product name or SKU) to narrow down the list. The system filters the database to show only the items that match your criteria.
3. Pagination: To keep the page performance high, the system displays products in manageable chunks. You can navigate through these pages using pagination controls.
4. Data Display: The system takes the raw product information from the database and translates it into a clear, user-friendly format, displaying key details like the product name, brand, SKU, code, tax, and price.
5. Empty State Handling: If no products match your search or if the catalog is empty, the system will gracefully inform you that no results were found rather than displaying an error.

2.3 Key Concepts

- Pagination: A method of breaking a large list of items into smaller, separate pages to improve loading speed and readability.
- Search Filtering: The ability to narrow down a large list of products by typing specific identifiers like a name or SKU.
- Data Mapping: The process of taking internal system data (like database field names) and converting them into clear, readable labels for the user interface.
- SKU (Stock Keeping Unit): A unique identifier used to track individual products within the inventory.

2.4 Business Value

- Improved Efficiency: Staff can quickly locate specific products without scrolling through long lists, saving time during daily operations.
- Scalability: By using pagination, the system remains responsive and easy to use regardless of how many thousands of products are added to the inventory.
- Data Clarity: Standardized labels ensure that all users, regardless of their technical background, can easily understand product details like pricing and tax information.
- Reliability: The system includes built-in checks to handle errors or empty searches, ensuring a smooth and professional user experience.

3. Data

3.1 Overview

The Product List Page feature provides a centralized view for users to browse and manage the company's inventory. This system retrieves product information from our database and organizes it into a clear, standardized format so that it can be displayed accurately on the screen. It is designed to be resilient, ensuring that even if data is missing or a connection issue occurs, the page remains stable for the user.

3.2 Step-by-Step Flow

1. Data Retrieval: The system queries the product database to gather all available inventory records.
2. Standardization: Each product record is processed to ensure the information matches the specific labels required for the user interface (such as SKU, Name, Brand, Product Code, Tax, and Price).
3. Delivery: The formatted list of products is sent to the display layer to be shown to the user.
4. Error Handling: If no products are found, the system gracefully displays an empty list. If a technical issue occurs, the system logs the error and returns an empty result to prevent the page from crashing.

3.3 Key Concepts

- Product Record: The raw information stored in our system for each item, including its unique identifier, pricing, and descriptive details.
- Data Mapping: The process of taking raw database information and translating it into the specific labels and formats used on the Product List Page.
- Resilience: The system's ability to handle missing information or connection errors by providing default values or empty states, ensuring a smooth user experience.

3.4 Business Value

This feature serves as the foundation for inventory management within the Mini ERP. By providing a reliable and organized list of products, it allows staff to quickly locate items, verify pricing and tax details, and maintain accurate records. The robust error handling ensures that the system remains dependable, reducing downtime and helping users maintain productivity even when data issues arise.

BUSINESS REQUIREMENTS DOCUMENT

Product Detail Page

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Frontend

1.1 Overview

The Product Detail Page is a comprehensive dashboard within the Mini ERP system designed to provide a centralized view of all information related to a specific product. It allows users to view detailed specifications, pricing, inventory status, and historical data in one place, while also providing quick access to management tools like editing, barcode generation, and tracking.

1.2 Step-by-Step Flow

1. Loading: When a user opens the page, the system automatically fetches the specific product's data from the database. A loading indicator is displayed until the information is ready.
2. Header & Identification: The page displays a persistent navigation bar and a "Product Chip" that clearly identifies the product by its name and SKU.
3. Action Toolbar: Users have immediate access to primary actions, including returning to the product list, editing the product details, or generating QR and barcode labels.
4. Data Organization: The page uses a two-column layout to organize information:

* Left Column: Focuses on core product data, including general details, tax information, various pricing tiers (Company, Store, and External), and tracking/bundle information.
* Right Column: Focuses on operational and administrative details, such as product attributes, dimensions, scheduling, settings, warranty, return policies, and trade-in information.

5. Deep-Dive Menus: Users can access specialized views for warehouse stock levels and historical purchase records through dedicated menu options.

1.3 Key Concepts

- Product Chip: A visual summary element that displays the product's unique identifier (SKU) and name for quick reference.
- Two-Column Grid: A structured layout design that separates core product data from administrative and operational settings to improve readability.
- State Management: The system’s ability to track whether data is currently loading, has successfully loaded, or has encountered an error, ensuring the user is always informed of the page status.
- Contextual Navigation: Buttons and menus that allow the user to perform specific tasks related to the product, such as editing or viewing stock, without leaving the current page.

1.4 Business Value

This feature serves as the "single source of truth" for any product in the ERP system. By consolidating complex data—ranging from pricing and tax rules to warranty and stock history—into a single, well-organized view, it

significantly reduces the time staff spend searching for information. The inclusion of direct action buttons (like Edit, QR, and Barcode) streamlines daily operational workflows, allowing for faster inventory management and administrative updates.

BUSINESS REQUIREMENTS DOCUMENT

Common Standard Response JSON Schema

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

This feature establishes a standardized "language" for how our software communicates its status back to users and other systems. By creating a consistent format for every type of response—whether a request was successful or encountered an error—we ensure that our API behaves predictably. This consistency makes it easier for developers to build reliable integrations and helps our system provide clear, actionable feedback when things go wrong.

1.2 Step-by-Step Flow

1. Registry Lookup: The system acts as a central registry that identifies the outcome of a request (e.g., success, unauthorized access, or server error).
2. Routing: Based on the specific outcome, the system routes the request to the corresponding "Response Template."
3. Template Application: The system applies the specific rules for that outcome (e.g., a 201 Created response will include the new data, while a 400 Bad Request will include specific error details).
4. Standardized Output: The final response is delivered in a uniform format, ensuring that the structure of the information remains consistent regardless of the specific status code.

1.3 Key Concepts

- Standardized Response: A uniform structure for all API communications, ensuring that every response follows the same layout.
- Status Codes: Numerical identifiers (like 201, 404, or 500) that tell the system and the user exactly what happened during a request.
- Schema: A set of rules that defines exactly what information must be included in a response (e.g., requiring an error message for failed requests).
- Validation Details: Specific information provided in error responses to help users understand exactly what went wrong (e.g., which field in a form was filled out incorrectly).

1.4 Business Value

- Improved Reliability: By enforcing a standard format, we reduce the likelihood of integration errors, as external systems will always know what to expect from our API.
- Faster Troubleshooting: Standardized error messages provide clear, consistent feedback, allowing support teams and developers to diagnose and resolve issues much faster.
- Better Developer Experience: A predictable API is easier to learn and use, which accelerates the development of new features and third-party integrations.
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System Stability: Consistent response handling ensures that the system manages errors gracefully, preventing unexpected crashes or confusing behavior for the end user.

BUSINESS REQUIREMENTS DOCUMENT

Store Stock History Schema

Prepared by Kolosal AI

Version	Date	Status	Classification
1.0	October 27, 2026	Draft	Internal

1. Data

1.1 Overview

The Store Stock History feature provides a structured way to track and audit inventory changes across different store locations. By maintaining a detailed log of every increase or decrease in stock, the system ensures that businesses can accurately monitor product movement, investigate discrepancies, and maintain reliable inventory records.

1.2 Step-by-Step Flow

1. Define Data Structures: The system establishes clear formats for three core areas: the current stock levels at each store, the history of changes made to that stock, and the store information itself.
2. Establish Response Format: The system defines how inventory history data is presented to users, including features like search, sorting, and pagination to handle large amounts of data efficiently.
3. Execute Data Queries: When a user requests information, the system performs a combined search across stores, current stock, and historical logs to provide a complete picture of inventory activity.
4. Map Data to Records: The system ensures that the information retrieved from the database is accurately translated into the defined formats for current stock, historical logs, and store details.
5. Calculate Totals: As part of the reporting process, the system automatically calculates the total number of items changed (net increase or decrease) to provide quick insights into stock trends.

1.3 Key Concepts

- Inventory Audit Log: A permanent record of every stock movement, including the reason for the change and the timestamp, which helps in tracking accountability.
- Data Mapping: The process of ensuring that information stored in the database matches the format required by the application, preventing errors and data inconsistencies.
- Relational Linking: The ability to connect different pieces of information—such as linking a specific product change to a specific store—to provide context to the data.
- Pagination and Filtering: Tools that allow users to navigate through large lists of inventory history by searching for specific dates, product IDs, or store names.

1.4 Business Value

- Improved Accuracy: By logging every stock change, the business reduces the risk of "ghost inventory" and ensures that the stock levels shown in the system match reality.
- Enhanced Accountability: With detailed history logs, managers can easily identify when and why stock levels changed, making it easier to spot errors or theft.
-

Better Decision Making: The ability to filter and summarize stock history allows leadership to identify trends, such as which products are moving fastest or which stores are experiencing frequent inventory adjustments.

- Operational Efficiency: Standardized data structures ensure that reporting is fast and reliable, saving time for staff who need to reconcile inventory or generate stock reports.

BUSINESS REQUIREMENTS DOCUMENT

Store Schema

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

This feature establishes the standardized way our system handles "Store" information. It ensures that whenever the system retrieves a list of stores, the data is organized consistently, follows a strict format, and is accurately pulled from our database. By defining these rules, we ensure that different parts of our software can reliably communicate and display store information without errors.

1.2 Step-by-Step Flow

1. Define the Output Format: The system establishes a template for how a list of stores should look when presented (e.g., ensuring every store has a unique ID and a name).
2. Retrieve Data: The system performs a secure, read-only search of the database to gather all available store records.
3. Apply Data Rules: The system references a formal "blueprint" (schema) that dictates exactly what information is required for a store to be considered valid.
4. Final Integration: The system combines the retrieved data with the blueprint to ensure the final information sent to the user perfectly matches the required format.

1.3 Key Concepts

- Schema: A formal blueprint or set of rules that defines the structure of data, ensuring that required fields (like a unique store ID) are always present.
- Serialization: The process of converting raw database information into a standardized format (JSON) that our software applications can easily read and display.
- Unique Identifier: A permanent, system-generated code assigned to each store to ensure it is never confused with another location.
- Read-Only Operation: A safety measure that allows the system to view store information without the risk of accidentally changing or deleting any records.

1.4 Business Value

- Data Consistency: By enforcing a strict format, we prevent errors where missing or malformed store information might cause the application to crash or display incorrect data.
- Reliability: Standardizing how we pull store records ensures that the information remains accurate and predictable across all areas of the ERP system.
- Scalability: With a clear blueprint for what a "Store" is, it becomes much easier for our technical team to add new features or integrate with other systems in the future without needing to rebuild the foundation.

BUSINESS REQUIREMENTS DOCUMENT

Purchase History JSON Schema

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

The Purchase History feature establishes a standardized digital record for every product procurement within our Mini ERP system. By defining a strict structure for this data, we ensure that every purchase entry—from the date and supplier to the final cost breakdown—is captured consistently. This creates a reliable foundation for tracking inventory costs and supplier performance.

1.2 Step-by-Step Flow

1. Data Definition: We define a formal blueprint (schema) that dictates exactly what information must be included for every purchase record, such as the product, quantity, pricing, and tax details.
2. Validation: Before any purchase data is accepted into the system, it is checked against this blueprint. If any required information is missing or formatted incorrectly, the system rejects the entry to prevent data errors.
3. Database Storage: Once validated, the information is organized into the Purchase History table, ensuring that all financial and logistical data is stored in a structured, permanent format for future reporting.

1.3 Key Concepts

- Schema: A set of rules that defines the structure of our data, ensuring that every purchase record contains the same necessary fields.
- Purchase Record: A single entry in our system that captures the full details of a specific transaction, including the product, supplier, and financial totals.
- Data Integrity: The process of ensuring that only complete and correctly formatted information is saved, which prevents "dirty data" from entering our reports.
- Financial Breakdown: The automated calculation of costs, including the subtotal (quantity times price), tax, and the final total, ensuring consistency across all records.

1.4 Business Value

- Accurate Reporting: By enforcing a strict structure, we ensure that financial reports regarding supplier spending and inventory costs are always based on complete and reliable data.
- Operational Efficiency: Standardizing how we record purchases reduces manual errors and makes it easier for different parts of the ERP system to communicate and share information.
- Audit Readiness: Having a consistent, validated history of all purchases makes it significantly easier to track spending patterns and verify transactions during internal or external audits.

BUSINESS REQUIREMENTS DOCUMENT

warehouse stock JSON mapping

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

1. Data

1.1 Overview

This feature establishes a standardized way for our Mini ERP system to organize and retrieve warehouse stock information. By defining a strict structure for how stock data is packaged, we ensure that information—such as product names, warehouse locations, and various inventory counts—is always consistent, accurate, and ready to be used by other parts of the application.

1.2 Step-by-Step Flow

1. Define the Data Structure: The system establishes a formal blueprint (schema) that dictates exactly what information must be included for every stock record, such as product identifiers, warehouse names, and specific quantity types.
2. Retrieve the Data: The system runs a query against the warehouse database to pull the requested stock information, ensuring that all eight required data points are collected for every item.
3. Validate and Prepare: The system verifies that the retrieved data matches the established blueprint (e.g., ensuring quantities are recorded as numbers) so that the information is clean and ready for the system to use.

1.3 Key Concepts

- Schema: A standardized blueprint that defines the required fields and data types, ensuring consistency across the system.
- Inventory Quantity: The total amount of a product currently held in a specific warehouse.
- Reserved Quantity: Stock that has been set aside for specific customer orders but has not yet left the warehouse.
- Consignment Quantity: Stock held at a warehouse that is owned by a third party but managed by us.
- Available Quantity: The actual amount of stock ready for new orders, calculated by accounting for reserved and consignment items.

1.4 Business Value

This feature provides a "single source of truth" for inventory levels. By enforcing a strict data structure, we reduce the risk of errors when reporting stock levels, prevent data mismatches between different parts of the ERP, and ensure that warehouse managers always have an accurate view of what is available to sell versus what is already spoken for.

BUSINESS REQUIREMENTS DOCUMENT

Product Schema

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

1. Data

1.1 Overview

The Product Schema feature provides a standardized way for our system to organize, retrieve, and display product information. It acts as a "blueprint" that ensures all product data—including basic details, brand information, product types, and stock levels across different warehouses and stores—is consistent and easy to access. Whether you are viewing a list of products or looking at the history of a specific item, this feature ensures the information is structured reliably.

1.2 Step-by-Step Flow

1. Defining the Structure: The system establishes standard formats (schemas) for product lists and individual product details, ensuring essential information like the SKU, name, and brand are always present.
2. Gathering Inventory Data: The system queries warehouse and store stock records to understand current inventory levels.
3. Consolidating Information: The system performs a "consolidated query," which pulls together the core product details, brand, type, and stock quantities into a single, unified view.
4. Enriching with Context: The system joins this consolidated data with specific records, such as brand descriptions, product categories, and store-specific stock counts.
5. Tracking History: The system maintains a history log for each product, recording changes to pricing, status, and other key attributes over time.
6. Delivering the Response: The final, organized data is presented in a consistent format, ready to be used by the ERP interface or other system modules.

1.3 Key Concepts

- Schema: A set of rules that defines how data must be organized, ensuring that every product record has the same fields and structure.
- Consolidated Query: A process that combines data from multiple sources (like warehouse stock, store stock, and brand info) into one complete record.
- Inventory Tracking: The ability to see exactly how much of a product is available in specific warehouses and retail stores.
- Product History: A chronological log of changes made to a product, such as price updates or status changes, which helps with auditing and tracking.
- Pagination: A method used when displaying lists of products to break them into manageable pages, improving system performance and user experience.

1.4 Business Value

- Data Consistency: By enforcing a strict structure, we eliminate errors caused by missing or mismatched information, ensuring that reports and inventory counts are always accurate.
- Operational Efficiency: Staff can quickly access a "single source of truth" for any product, reducing the time spent searching through different systems to find stock levels or pricing history.
- Better Decision Making: With real-time visibility into stock across all locations and a clear history of product changes, management can make more informed decisions regarding purchasing, pricing, and inventory management.
- Scalability: The standardized schema allows the system to grow easily, making it simple to add new product types or warehouse locations without disrupting existing operations.

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Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

BUSINESS REQUIREMENTS DOCUMENT

Customer New Vehicle Category Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Filter and Download Section

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Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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POST /api/service_products/

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Product Detail Menu

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Sales Menu Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Sales Return Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Vehicle Categories List Page

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Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Vehicle Edit Page

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

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Customer Edit Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Vehicle Detail Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer New Vehicle Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Vehicles List Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Detail - Returns

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Detail - Purchases

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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New Customer Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer List Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Customer Detail Page

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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GET /api/service_products/inactive

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Details Card

Prepared by Kolosal AI

Version 1.0 **Date** October 27, 2026 **Status** Draft **Classification** Internal

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GET /api/service_products

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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GET /api/service_products?utf8= &q[attribute_predicate]=value&...&page_rows=

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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GET /api/service_products/new

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Tax Card

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Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Store Pricing Card

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal

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Settings Card

Prepared by Kolosal AI

Version 1.0	**Date** October 27, 2026	**Status** Draft	**Classification** Internal